

Higher-quality CAD across all tiers + CAD vs 3-D, made distinct v0.93.0

The CAD renderer got a quality pass on every tier; the rotating CAD and the 3-D model now read as different things.

THE QUALITY PASS — engine/cad_render.py (CAD_VERSION 4)

Anti-aliasing

Supersample 2x → 3x.
Cleaner edges everywhere.

Finer mesh

cyl 28→48 · tube 32→52 ·
torus 30×16→48×26 · sphere
16×12→28×20 · helix sv 8→12.
Round parts look round.

CAD ink-line

Crisp silhouette + hole
outline from the part mask —
the technical-drawing line.

Depth + clean faces

Soft contact shadow under
the part; facet edges = a
subtle tint, not a black wire.

Applies to ALL tiers, ladder preserved: v1 flat (legacy) · v2 +specular/metallic (lite) · v3 +FLIS colour/texture (modern). Proof: docs/cad_quality_v4.png.

THE TIER LADDER (each now higher quality)

LEGACY · v1

Flat diffuse, head-down, no colour/texture — but now SS3, rounder mesh, silhouette.

LITE · v2

Everything in v1 + right-side-up + specular / metallic highlights.

MODERN · v3

Everything in v2 + FLIS colour + per-material procedural surface texture.

CAD vs 3-D — now visually DISTINCT (same smooth WebGL motion)

📐 ROTATE CAD (technical)

- Flat-faceted, neutral machined STEEL, matte.
- GLV.load(geom, '#9aa6b2', false, [0.18,14,0.30]).
- Cool blueprint-grey stage. Reads as a CAD model.
- Carries shape + FLIS dimensions, not the part colour.

📐 INTERACTIVE 3-D (realistic)

- The part's FLIS colour, smooth shading.
- The scanned MATERIAL finish (metallic/matte/rubber).
- Looks like the real object's surface.
- Unchanged from before.

VERIFIED (host-side)

- VERIFY-CAD-QUALITY.bat: cad_render.py parses, both UI pages' inline JS parse (exit 0).
- All-tiers quality grid renders to docs/cad_quality_v4.png — round parts round, crisp silhouettes, soft shadows.
- Re-render the full library with RUN-CAD-TIERS.bat (clears + re-renders v1/v2/v3) to push v4 quality everywhere.
- Additive & rollbackable (R1): revert the render edits + the cadspin GLV.load line + CAD_VERSION.